Pressure pad power budget

Team Number:	203
Project Name:	**Pressure Pad Subsystem**
Team Member Names:	**Ramon Duarte**
Version:	**2.0 - 9V Motor Update**

All Major Components	**Component Name**	**Part Number**	**Supply**	**#**	**Absolute**	**Total**	**Unit**
	Microcontroller	PIC18F57Q43	1.8 - 5.5 V	1	250.00	250.00	mA
	Quad op-amp	MCP6004	1.8 - 6.0 V	1	0.68	0.68	mA
	Indicator LED	D1 (generic LED)	5 V rail (330 ohm series R)	1	20.00	20.00	mA
	Pressure pad / force sensor	FSR01CE	5 V circuit	1	0.50	0.50	mA
	H-bridge motor driver	SN754410NE	VCC1: 4.5-5.5 V; VCC2:	1	70.00	70.00	mA
	DC motor	RF370CA-15370	3 - 12 V operating range	1	795.00	795.00	mA
+9V Motor Power Rail	**Component Name**	**Part Number**	**Supply**	**#**	**Maximum /**	**Total**	**Unit**
	DC motor	RF370CA-15370	3 - 12 V operating range	1	795.00	795.00	mA
	H-bridge output supply	SN754410NE VCC2	4.5 - 36 V	1	0.00	0.00	mA
						0.00	mA
						0.00	mA
						0.00	mA
					Subtotal	795.00	mA
					Safety Margin	25%	
					Total Current Required on +9V Rail	993.75	mA
c1. Regulator or Source	+9V External Supply	Tensility 16-00134	9 VDC output	1	2000.00	2000.00	mA
					Total Remaining Current Available on +9V Rail	1006.25	mA
+5V Power Rail	**Component Name**	**Part Number**	**Supply**	**#**	**Absolute**	**Total**	**Unit**
	Microcontroller	PIC18F57Q43	1.8 - 5.5 V	1	250.00	250.00	mA
	Quad op-amp	MCP6004	1.8 - 6.0 V	1	0.68	0.68	mA
	Indicator LED	D1 (generic LED)	5 V rail (330 ohm series R)	1	20.00	20.00	mA
	Pressure pad / force sensor	FSR01CE	5 V circuit	1	0.50	0.50	mA
	H-bridge logic supply	SN754410NE VCC1	4.5 - 5.5 V	1	70.00	70.00	mA
					Subtotal	341.18	mA
					Safety Margin	25%	
					Total Current Required on +5V Rail	426.48	mA
c2. Regulator or Source	+5V Regulator	MC7805CTG	7 - 35 V input	1	1000.00	1000.00	mA
					Total Remaining Current Available on +5V Rail	573.53	mA
External Power Source 1	**Component Name**	**Part Number**	**Supply**	**Output Voltage**	**Absolute**	**Total**	**Unit**
External Power Source 1	Component Name	Part Number	Supply	Output Voltage	Absolute	Total	Unit
Power Source 1 Selection	AC/DC wall adapter	Tensility 16-00134	100-240 VAC	9 VDC	2000.00	2000.00	mA
						0.00	
	+5V Regulator + +9V Motor Rail	MC7805CTG +	9 V input	1	2000.00	2000.00	mA
						0.00	
	Estimated worst-case current					**1420.23**	**mA**
	Total Remaining Current Available					**579.77**	**mA**